

a) A straightforward way to find out if test subjects intuitively interpret darker or lighter colors as representing higher values would be to ask them to interpret plots without a legend. Why do the authors instead decide to show legends and evaluate response times? (2P)

Answer: We found two main reasons in the paper for this decision. First, if participants see colormaps without a legend, they might try to give the same answers in every trial to look consistent, even if the background changes. This would hide their real, natural reactions. Second, in real life, data visualizations almost always have legends. We think the authors wanted to test the colormaps in a natural, realistic way. They chose to measure response times because people generally answer much faster when the legend perfectly matches their natural feelings (their inferred mapping).

b) When interpreting visualizations, accuracy should be at least as important as response times. How do the authors account for that in their experiments? (2P)

Answer: We can see that the authors cared a lot about accuracy during both the experiment and the data analysis. During the experiment, they told participants to answer as quickly as possible but also to stay accurate. If participants made a mistake, they heard a warning sound. When analyzing the results, the authors first deleted all the trials where participants gave the wrong answer. In the second experiment, they were even stricter and completely removed the data of any participants who had an overall accuracy score below 90%.

c) Authors observe that neither the hypothesis of a “dark-is-more” bias nor a “contrast-is-more” bias could convincingly explain all available experimental results. What explanation do they propose instead? (2P)

Answer: The authors proposed a new idea called the "opaque-is-more" bias. This means that people naturally think colors that look more solid (more opaque or less transparent) represent bigger numbers. They explained that when a colormap looks like it has different levels of transparency, this new opaque-is-more bias can work together with the dark-is-more bias, or it can fight against it, depending on the background color.

d) Do all experimental results in the paper agree with their proposed explanation? If not, what is the exception and what do they mention as a potential reason for it? (2P)

Answer: No, not all results perfectly agree with their idea. We found an exception in Experiment 2 with the blue-white color scale on a black background. Here, the dark-is-more bias disappeared, even though the colors did not look like they had different opacity levels. The authors think this happened because of "carryover effects." Because participants saw other colormaps that really had opacity variations on the same black background during the test, they probably carried that feeling over to this blue-white colormap too.

e) Authors propose an “Opacity Variation Index” whose definition relies on the CIELAB color space. Do you see a reason for preferring CIELAB over alternatives such as RGB or XYZ for this purpose? (2P)

Answer: We think the main reason is that CIELAB is a "device-independent" color space. This means the colors do not change depending on the computer monitor, unlike RGB, which is device-dependent. Also, CIELAB represents human color perception much better. It has "metric properties," which means the mathematical distance between points in the CIELAB space actually matches how our human eyes see the difference between colors. This makes it the best choice for calculating a visual index like the Opacity Variation Index.

f) Several plots include error bars that represent standard errors of the means. Please explain how that differs from the standard deviation of the measurements. Use additional sources of information if needed. (2P)

Answer: Standard deviation tells us how much the individual data points in our group spread out from the average (mean). It shows the natural variation in our specific sample. On the other hand, the standard error of the mean tells us how accurate our sample's average is compared to the real average of the whole population. The standard error is smaller because we calculate it by dividing the standard deviation by the square root of our sample size. We use the standard error in our graphs to show how confident we are about the average value we found.